

Supervised Learning } applicable for labelled dataset
dataset with ground truth

Regression
Label: Continuous ND

Classification
Label: Categorical
or
Discrete ND

Inferential Stat modelling process

Assuming the given dataset is COMPATIBLE for ML algo.

- ① Data must be COMPLETE
- ② Data must be STRICTLY NUMERIC

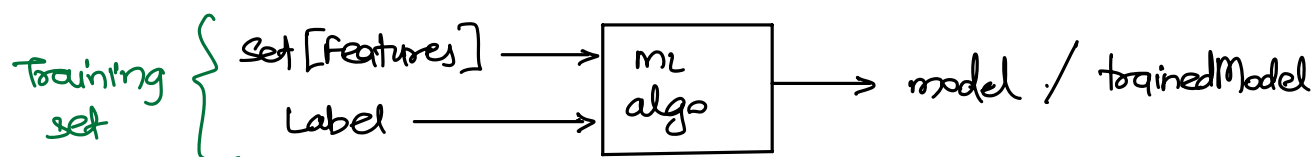
① Separate data as features and label (represent the data in the form of numpy array)

② Split the data into Training set and Testing set.

Training set → used to train the model (Data Convergence)

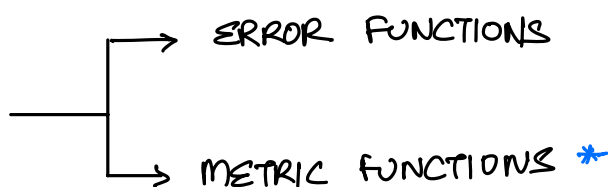
Testing set → used to check the quality of the model.

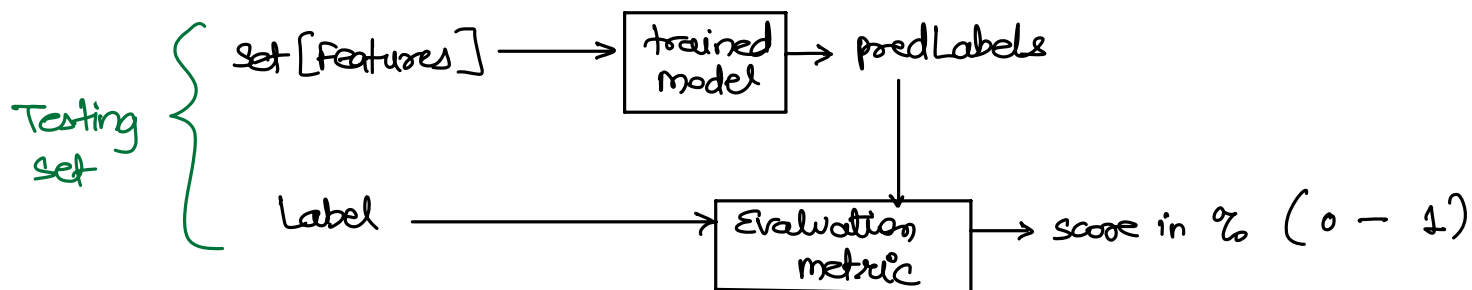
③ Select the relevant ML algorithms and start the training process.



④ Check the quality of the model / model's prediction.

Goal: check the quality of the model w.r.t prediction





⑤ Decide whether to Approve model or Improve model.

if eval passed:

Approve the model and deploy the same

else:

Improve model and recheck the quality.

Regression Algorithms

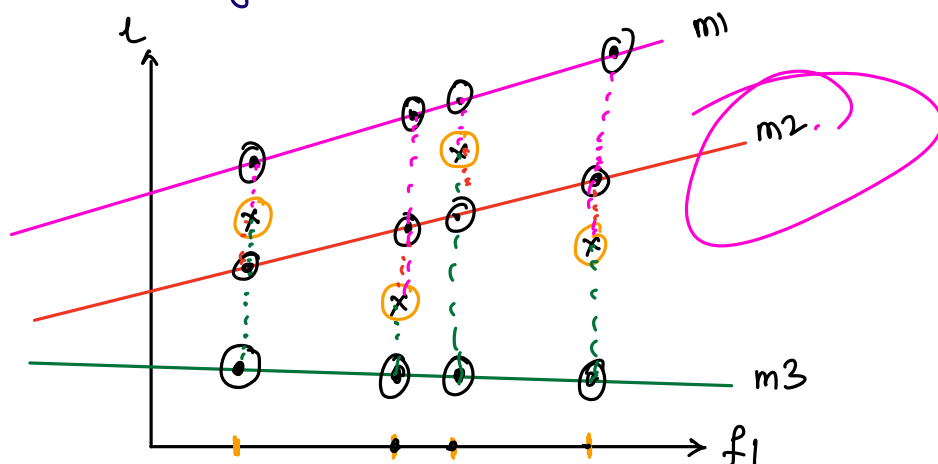
label: if ground truth is continuous ND, choose REGRESSION algo.

Linear Regression

This algo is based on SLOPE INTERCEPT FORMULA.

$$y = mx + c$$

Goal: To identify the LINE OF BEST FIT



Error: difference between actual label & predicted label.

create a model that can predict the profit of the company based on company's spending pattern and company's loc.

→ California Florida NY odSpend admSpend markSpend

ℳ → profit

label → $y = b_0 + \underbrace{b_1 x_1}_{\text{coeff}} + \underbrace{b_2 x_2}_{\text{coeff}} + \dots + \underbrace{b_n x_n}_{\text{coeff}}$

Annotations:
- b_0 is labeled "y intercept" (blue).
- $b_1 x_1$, $b_2 x_2$, and $b_n x_n$ are collectively labeled "feature column data" (green).
- b_1 , b_2 , and b_n are labeled "coeff" (orange).

$$\text{profit} = b_0 + b_1 (\text{California}) + b_2 (\text{Florida}) + b_3 (\text{NY}) + b_4 (\text{odSpend}) + b_5 (\text{admSpend}) + b_6 (\text{markSpend})$$